

CSE_2323

Shift Register

Digital Logic Design

Course code: CSE-2323

Credit Hour: 3

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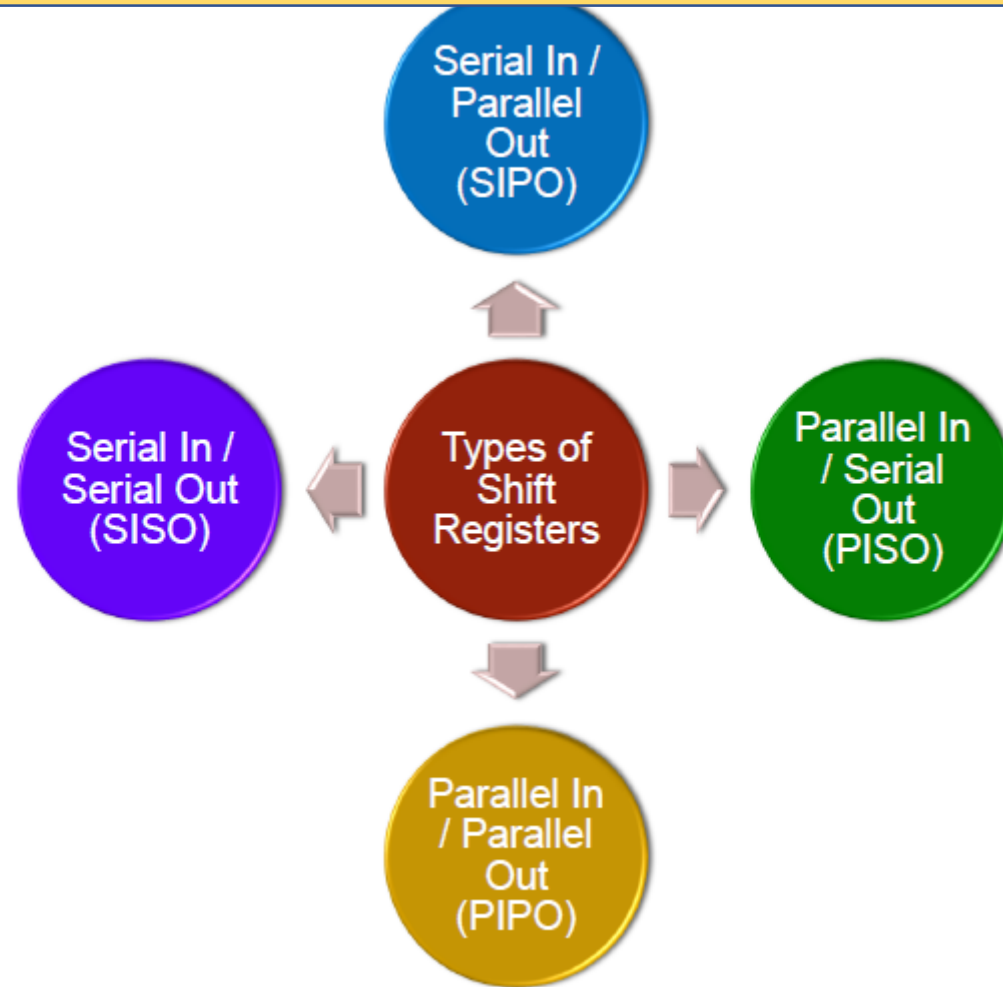
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Shift Register

- In Shift Registers, the binary bits transfers (shifts) from the output of one flip-flop to the input of the next individual Flip-Flop at every clock edge.
- Shift Registers are an important Flip-Flop configuration with a wide range of applications, including:
 - Computer and Data Communications
 - Serial and Parallel Communications
 - Multi-bit number storage
 - Sequencing
 - Basic arithmetic such as scaling (a serial shift to the left or right will change the value of a binary number a power of 2)
 - Logical operations

Types of shift register

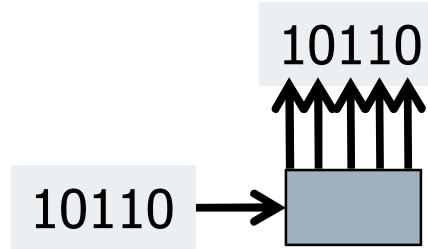


Types of shift register

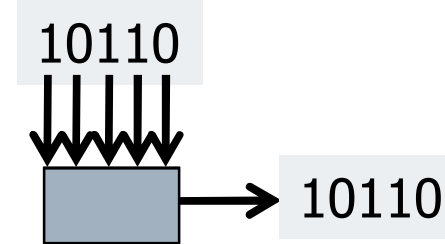
- SISO: Serial In, Serial Out



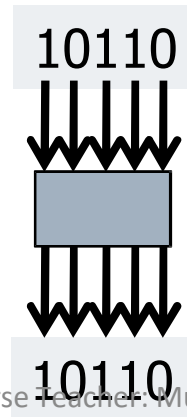
- SIPO: Serial In, Parallel Out



- PISO: Parallel In, Serial Out

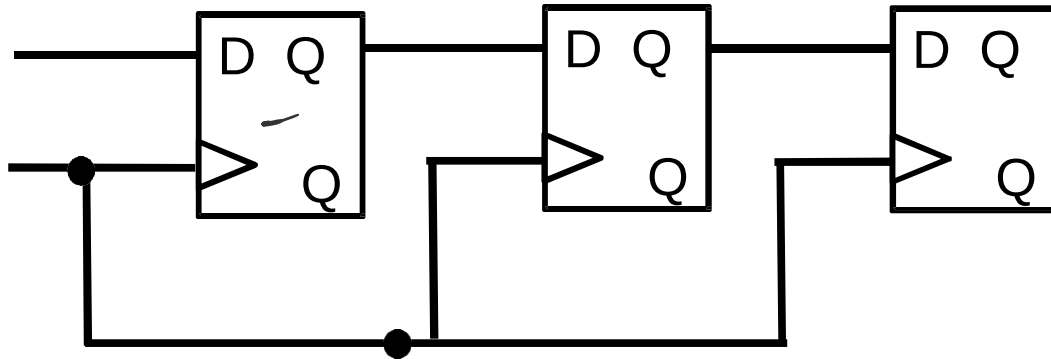


- PIPO: Parallel In, Parallel Out



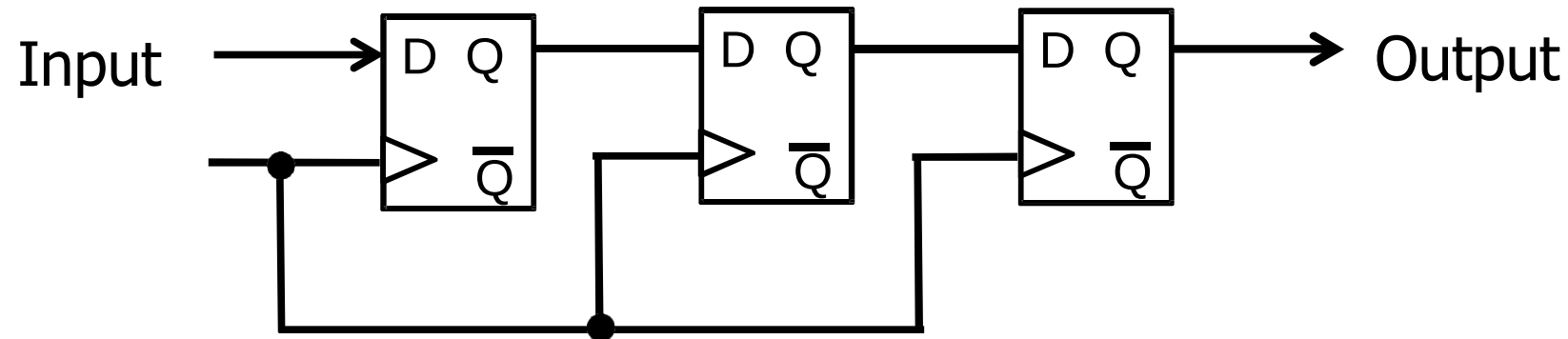
Shift Register Construction

- Shift registers are comprised of D Flip-Flops that share a common clock input.



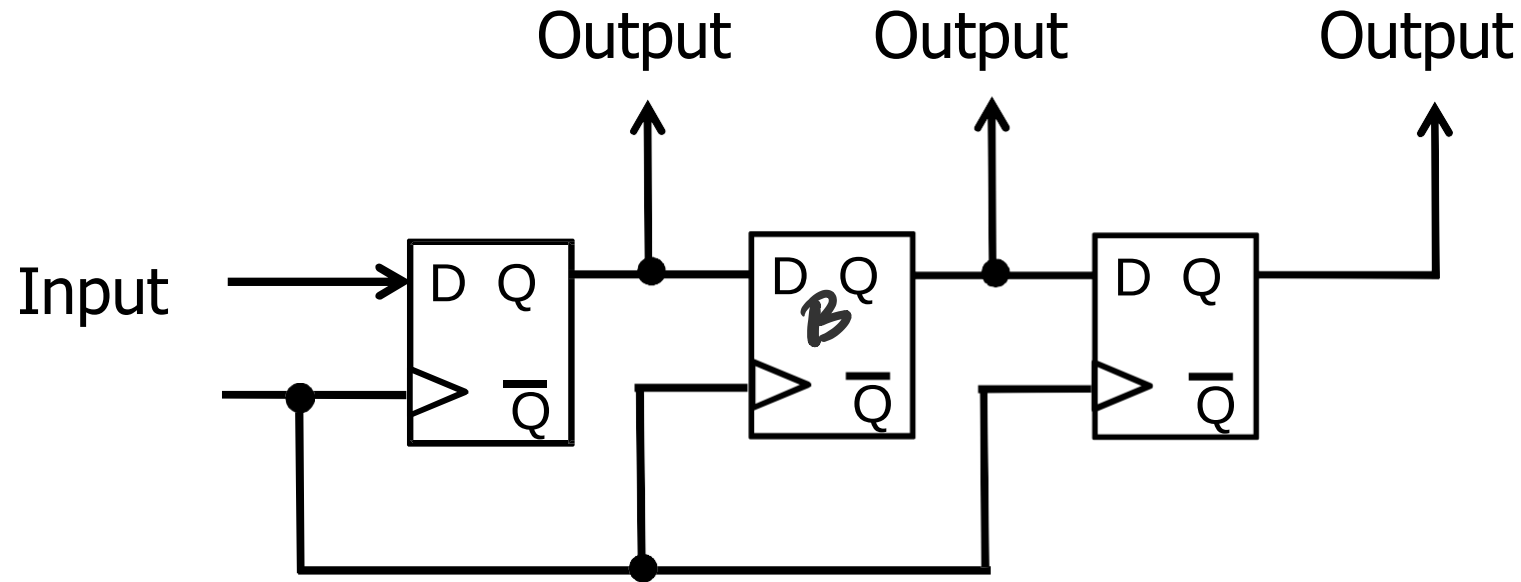
SISO Flip-Flop Shift Register

- a Serial In Serial Out shift register has a single input and a single output



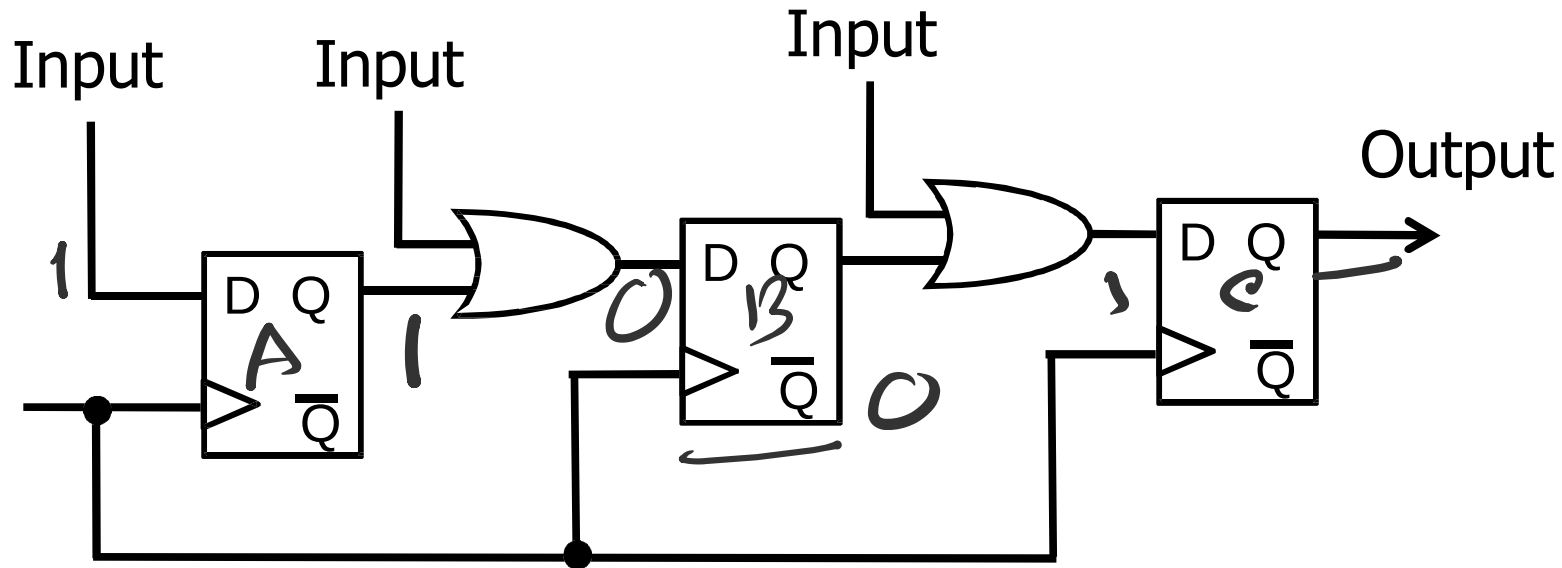
SISO Flip-Flop Shift Register

- a Serial In Parallel Out shift register has a single input and access to all outputs



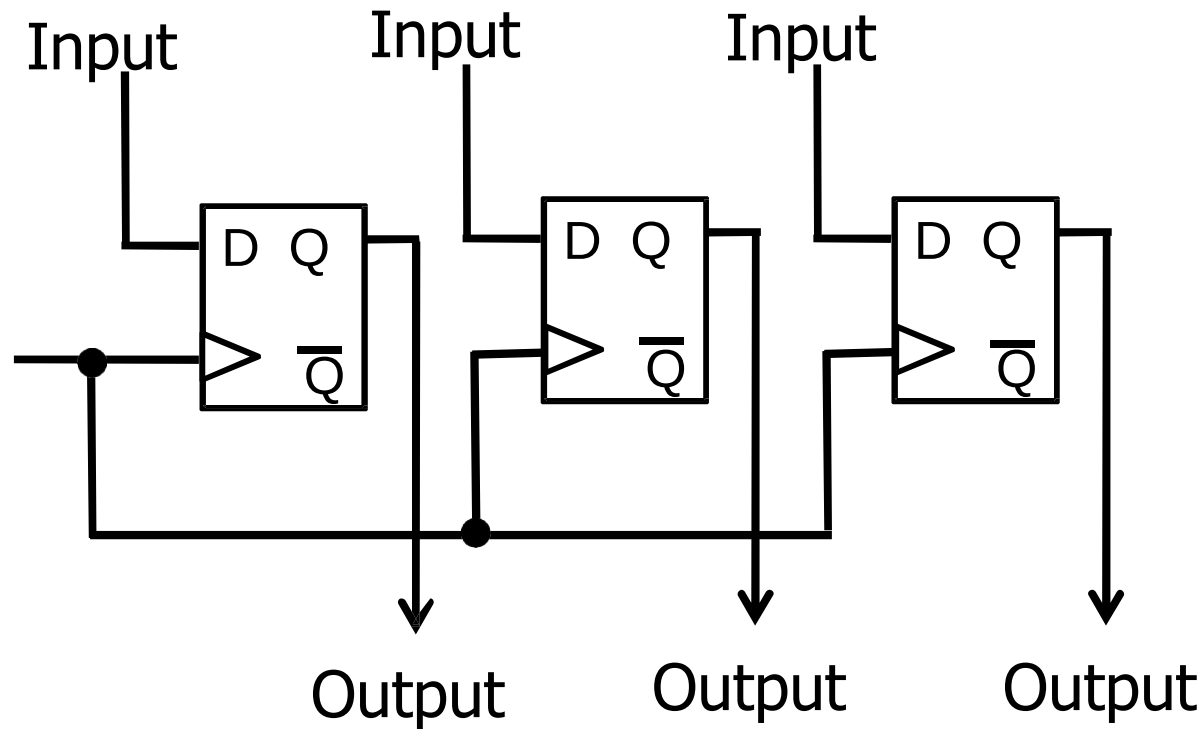
PISO Flip-Flop Shift Register

- a *Parallel In Serial Out* shift register requires additional gates, and the parallel input must revert to logic low.



PIPO Flip-Flop Shift Register

- a *Parallel In Parallel Out* register has the simplest configuration. It represents a memory device.



Universal Shift Registers

- Universal Shift Registers can be configured to operate in a variety of modes.
- For instance, they can be configured to have either Serial or Parallel Input/Output

A universal shift register operates in any of three modes:

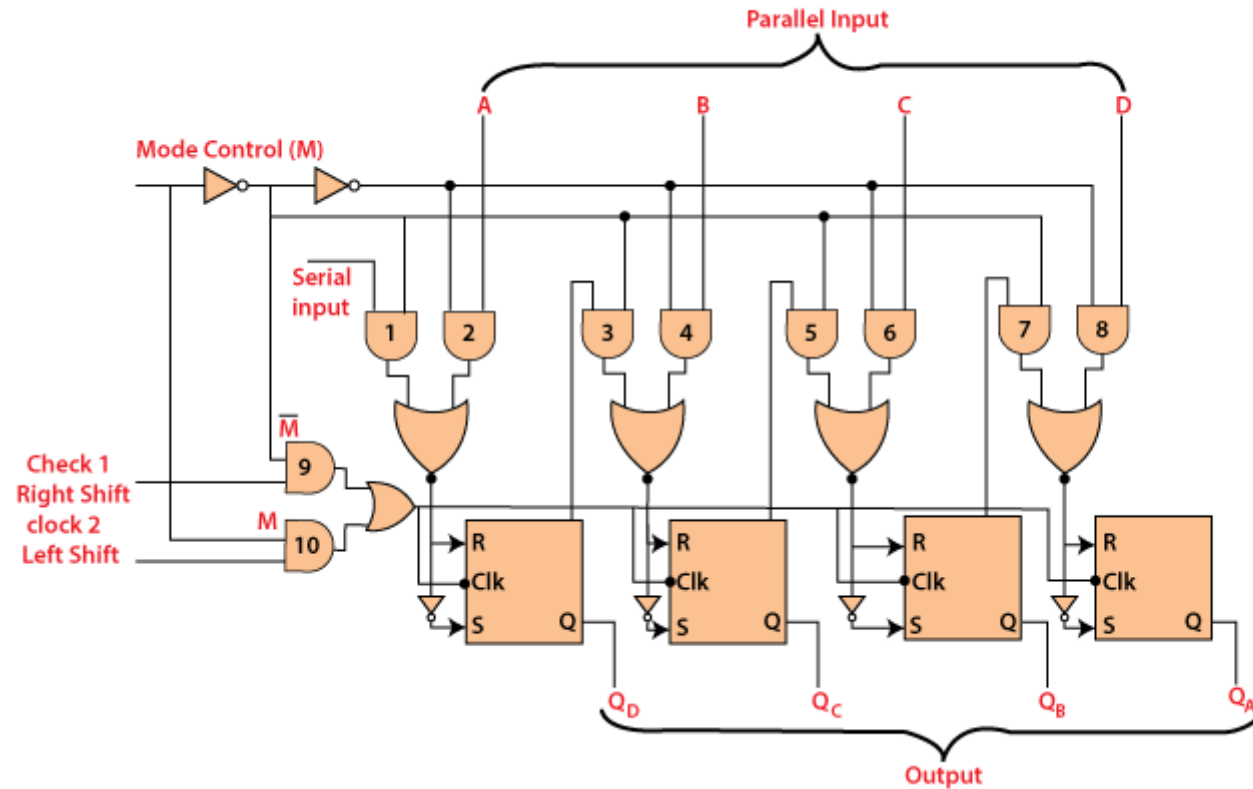
If the "Shift Left" control signal is active, there is a shift in data one position to the left, while the leftmost bit is lost and the rightmost bit is replaced from the serial input.

In "Shift Right," the data moves right with the loss of the rightmost bit and loading of the leftmost from the serial input.

"Parallel Load" allows the simultaneous loading of data from all input lines. For this, the register retains its current data if none of the control signals have been activated. Internally, their obvious external interface is implemented by using multiplexers and flip-flops: the bits are held by a flip-flop, while a multiplexer acts for the input source at each operation

Universal Shift Registers

Block Diagram



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RAM and ROM

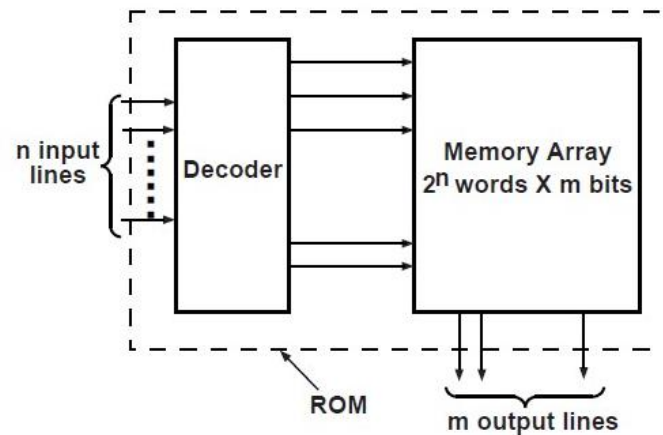
Primary Memory

- It is the only memory that is directly accessible to CPU.
- The CPU continuously reads instructions stored there and executes them as required.
- Various "Primary storage devices" are:
 - RAM
 - ROM
 - Cache Memory

ROM

A ROM is a memory (or storage) device in which permanent binary information is stored.

Once the pattern is established, it stays within the unit even when power is turned off and on again



A read-only memory (ROM) is a device that includes both the decoder and the OR gates within a single IC package. The connections between the outputs of the decoder and the inputs of the OR gates can be specified for each particular configuration

ROM

It consists of n input lines and m output lines.

Each bit combination of the input variables is called an address.

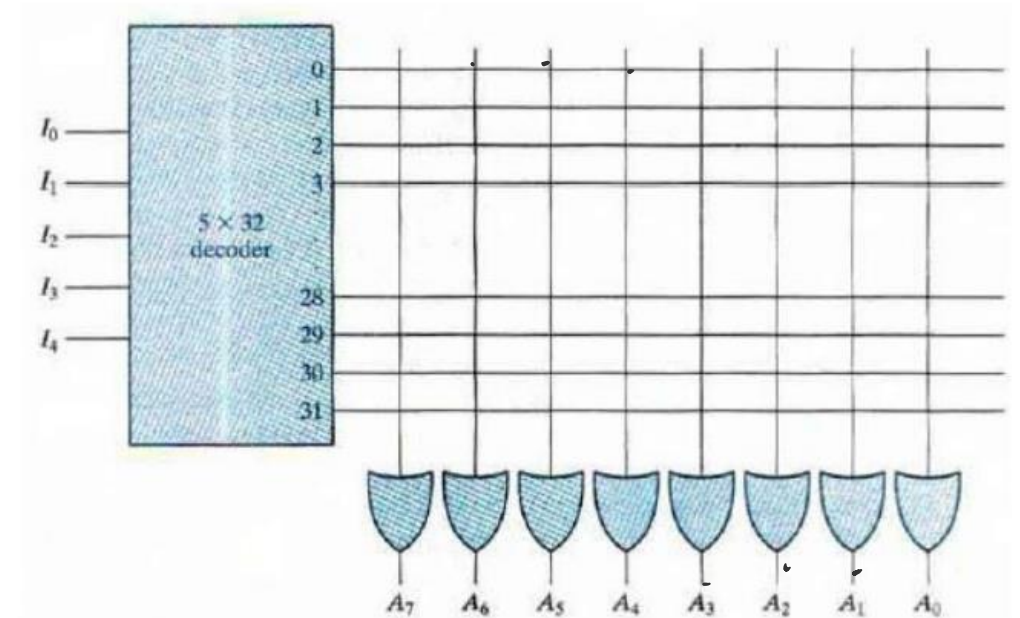
Each bit combination that comes out of the output lines is called a word,

The number of bits per word is equal to the number of output lines, m

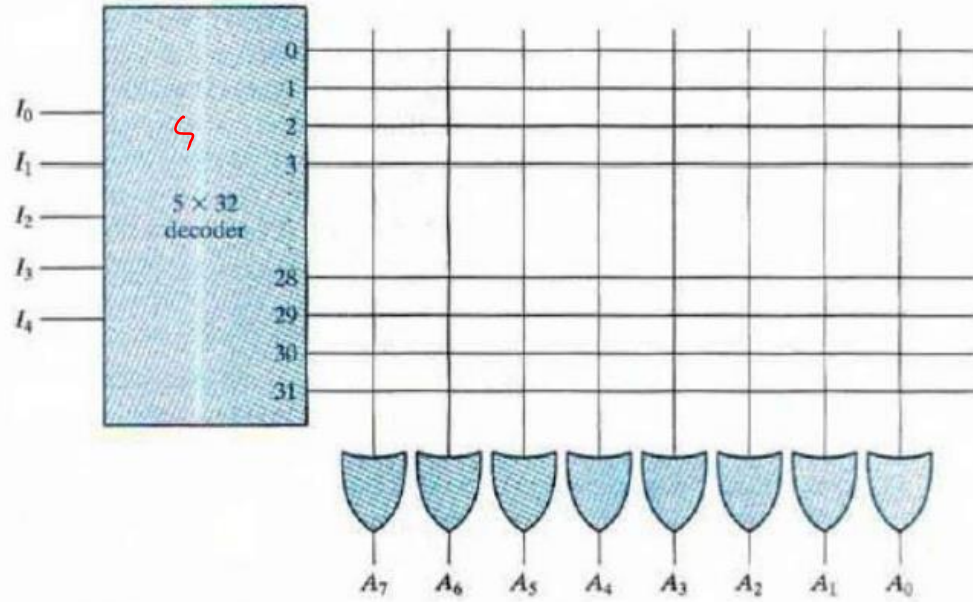
The number of distinct addresses possible with n input variables is 2^n

The word available on the output lines at any given time depends on the address value applied to the input lines.

A ROM is characterized by the number of words 2^n and the number of bits per word m .

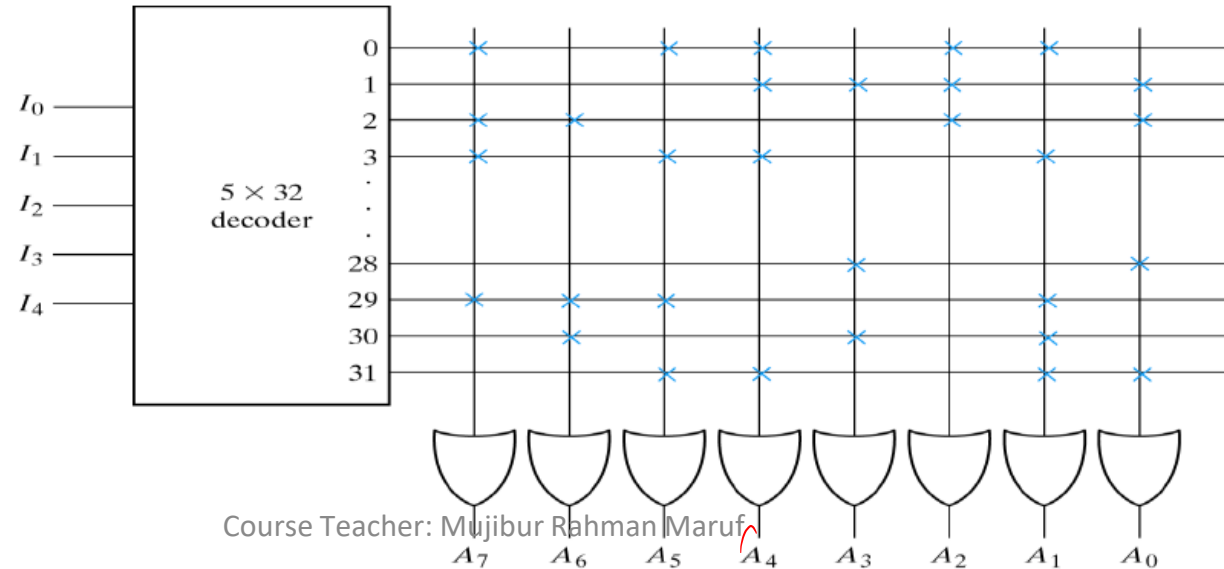


ROM



ROM Truth Table (Partial)

Inputs					Outputs							
I_4	I_3	I_2	I_1	I_0	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
0	0	0	0	0	1	0	1	1	0	1	1	0
0	0	0	0	1	0	0	0	1	1	1	0	1
0	0	0	1	0	1	1	0	0	0	1	0	1
0	0	0	1	1	1	0	1	1	0	0	1	0
		$\vdots$						$\vdots$				
1	1	1	0	0	0	0	0	0	1	0	0	1
1	1	1	0	1	1	1	1	0	0	0	1	0
1	1	1	1	0	0	1	0	0	1	0	1	0
1	1	1	1	1	0	0	1	1	0	0	1	1



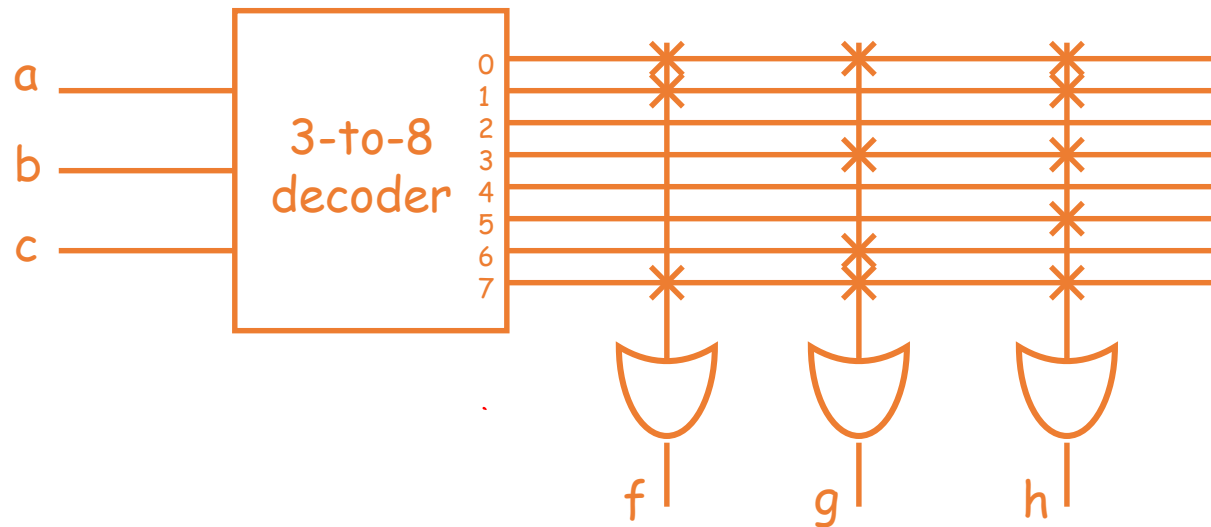
Example:

Implement the function with ROM.

$$f = \sum m(0, 1, 7)$$

$$g = \sum m(0, 3, 6, 7)$$

$$h = \sum m(0, 1, 3, 5, 7)$$



Example:

Implement the following function with ROM

$$F(w,x,y,z)=\Sigma(0,1,3,4,8,9,15)$$

Types of ROM

■ Mask ROM (MROM)

- This type of read-only memory (ROM) whose contents are programmed by the integrated circuit manufacturer (rather than by the user).
- User cannot erase content and they are not allowed to write new contents in masked ROM.

PROM (Programmable Read Only Memory)

- On Programmable Read Only Memory the Data can be written only once.
- It is Manufactured as a blank memory and then the data is written on it , also called as **Burning the PROM**.
- The process of writing your data to the PROM involves a special piece of equipment called a **PROM burner**. ✓
- The device programmer writes data to the device one word at a time by applying an electrical charge to the input pins of the chip. Once a PROM has been programmed in this way, its contents can never be changed.

Types of ROM

EPROM (Erasable Programmable read Only Memory)

- An EPROM (erasable-and-programmable ROM) is programmed in exactly the same manner as a PROM. However, EPROMs can be erased and reprogrammed repeatedly.
- To erase an EPROM, you simply expose the device to a strong source of ultraviolet light. (A window in the top of the device allows the light to reach the silicon.)

EEPROM (Electrically Erasable Programmable Read Only Memory)

- EEPROM is a special type of PROM that can be erased by exposing it to an electric charge.
- EEPROMS are electrically-erasable-and-programmable. Internally, they are similar to EPROMs, but the erase operation is accomplished electrically, rather than by exposure to ultraviolet light.
- Any byte within an EEPROM may be erased and rewritten. Once written, the new data will remain in the device forever- or at least until it is electrically erased.

Types of ROM

Flash memory:

Flash memory is an electronic non-volatile computer memory storage medium that can be electrically erased and reprogrammed

Example: memory card, pendrive

RAM(Random Access Memory)

- Random Access Memory allows stored data to be accessed directly by CPU in any random order.
- It is so called random access as data and instructions can read from or written into RAM in random order.
- It stores temporary data of currently running applications and play vital role in computers operation.
- It is a Volatile Memory i.e. the data vanishes/erased when the power supply is cut off.

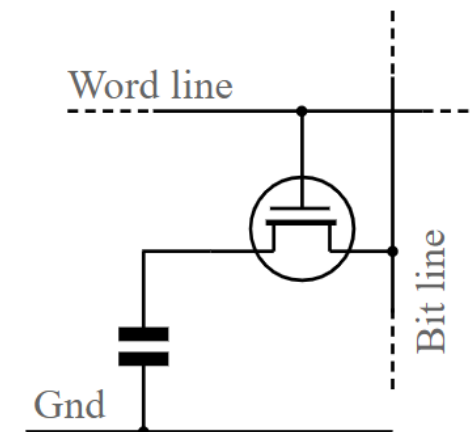
Two types

DRAM (Dynamic random access memory)

SRAM (Static Random Access Memory)

DRAM (Dynamic random access memory)

- This is a Random Access Memory that stores each bit of data in a separate capacitor. ✓
- The capacitor can be either charged or discharged; these two states are taken to represent the two values of bit, that are 0 and 1. S
- Main characteristics of capacitors is that they leaks electric charge gradually. This will result in turning back the 1 state gradually to 0 state i.e. data may get lost in long run.

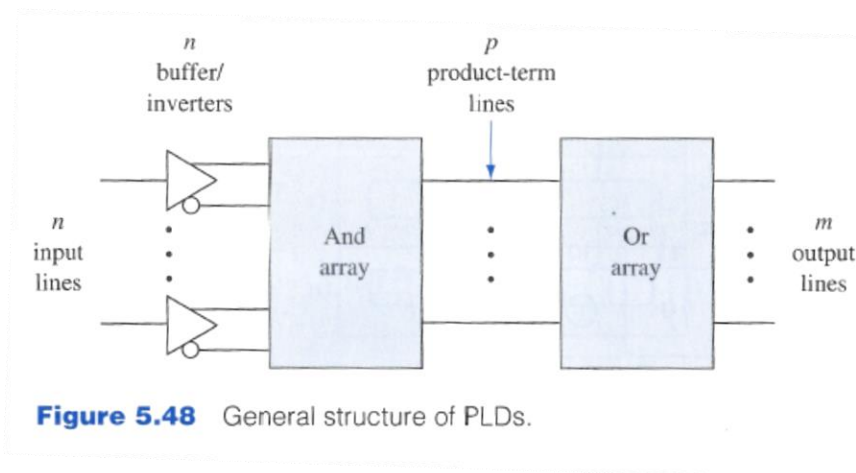


SRAM (Static Random Access Memory)

- **Static random-access memory** (static RAM or SRAM) is a type of semiconductor memory that uses bistable latching circuitry (flip-flop) and transistors to store each bit.
- **Volatile** in the conventional sense that data is eventually lost when the memory is not powered.
- It is faster than the DRAM as it uses a six transistor configuration.

General Structure of PLD

- Inputs to the PLD are applied to a set of buffer/inverters. These devices have both the true value of the input as well as the complemented value of the input as its outputs.
- Outputs from these devices are the inputs to an array of and-gates. The AND array generates a set of p product terms.
- The product terms are inputs to an array of or-gates to realize a set of m sum-of-product expressions.



General Structure of PLD

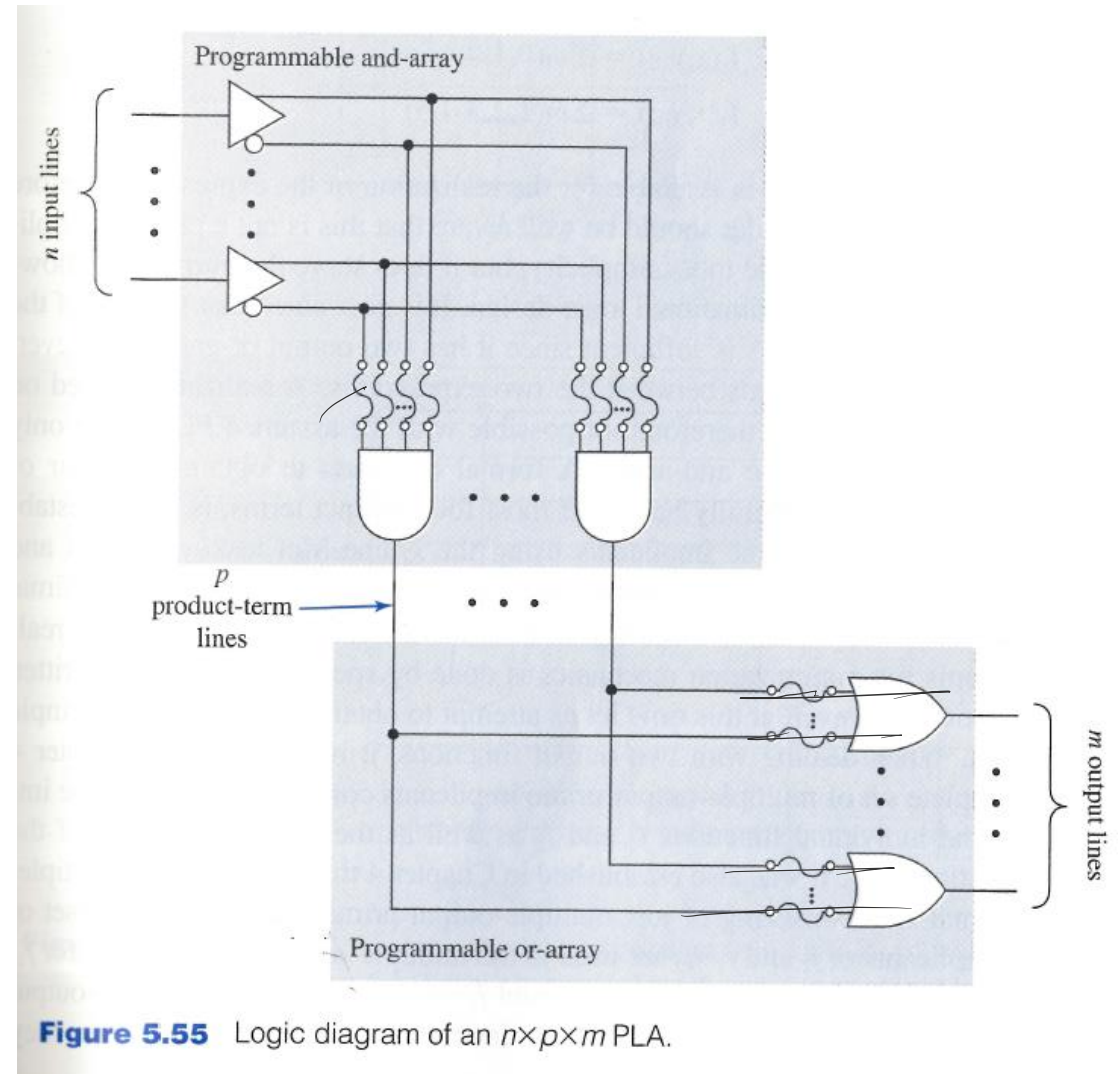
- One or both of the gate arrays are programmable.
- The logic designer can specify the connections within an array.
- PLDs serve as general circuits for the realization of a set of Boolean functions.

Device	AND-array	OR-array
PROM	Fixed	Programmable
PLA	Programmable	Programmable
PAL	Programmable	Fixed

Programmable Logic Array

- PLAs are characterized by three numbers:
 - Number of input lines n
 - Number of product terms that can be generated p (the number of AND gates)
 - Number of output lines m
- $n \times p \times m$ PLAs
- Typical PLA is $16 \times 48 \times 8$.

PAZ



Programmable Array Logic (PAL) Devices

- OR-array is fixed by the manufacturer of the device.
 - PAL device is easier to program and less expensive than the PLA.
 - Less flexible.

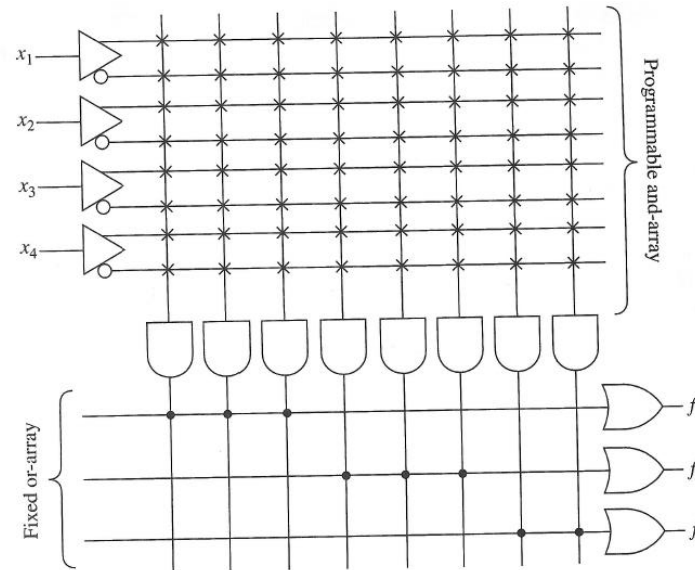


Figure 5.62 A simple four-input, three-output PAL device.

For our examples:

- 4-input, 3-output PAL device
- Three Boolean expressions can be realized in which *two expressions can have at most 3 product terms and one expression can have at most 2 product terms.*

Example:

A. Implement the circuit with PLA

B. Implement the circuit with PAL

$$f = \sum m(0, 1, 7)$$

$$g = \sum m(0, 3, 6, 7)$$

$$h = \sum m(0, 1, 3, 5, 7)$$

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Finite State Machine

Finite State Machine

- It is an abstract machine that **can be in exactly one of a finite number of states at any** given time.

Combinational Circuit

- Output Depends on present input

Sequential Circuit

- Output depends on present input and present state

Two types of finite state machines

- Mealy machine

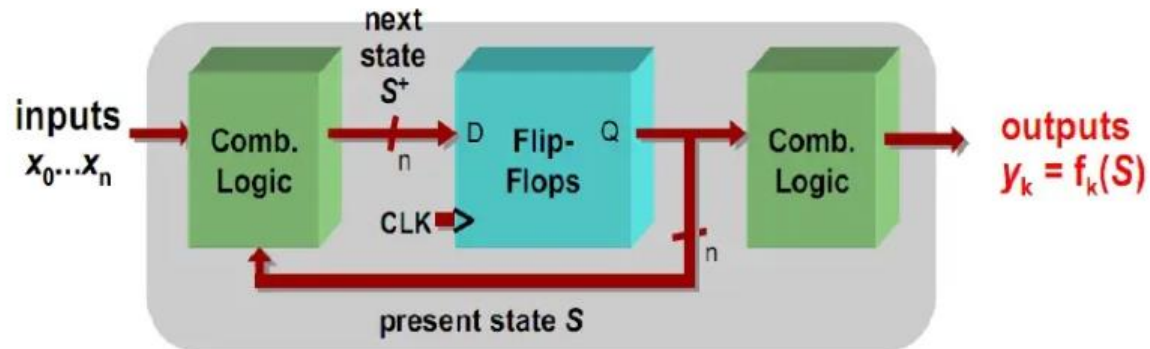
- Output is function of present state and present input

- Moore machine

- Output is function of present state only

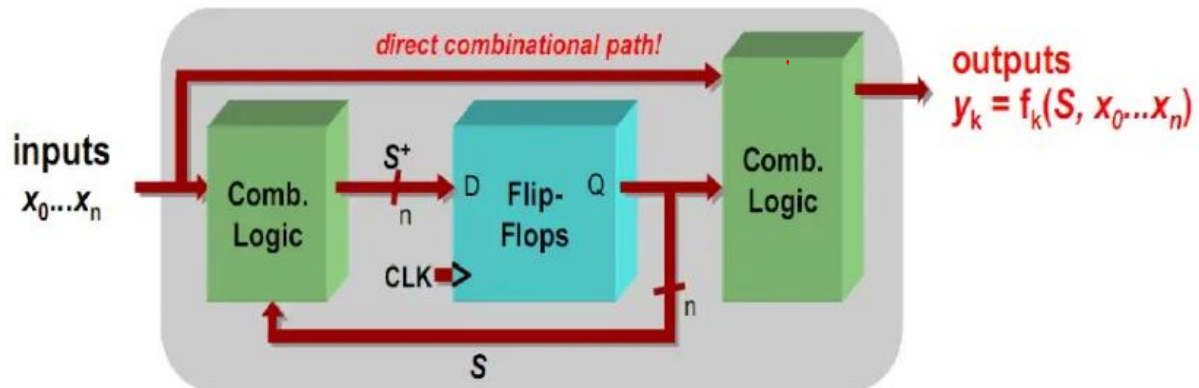
Design example of mealy Machine

MOORE STATE MACHINE



Moore State Machine

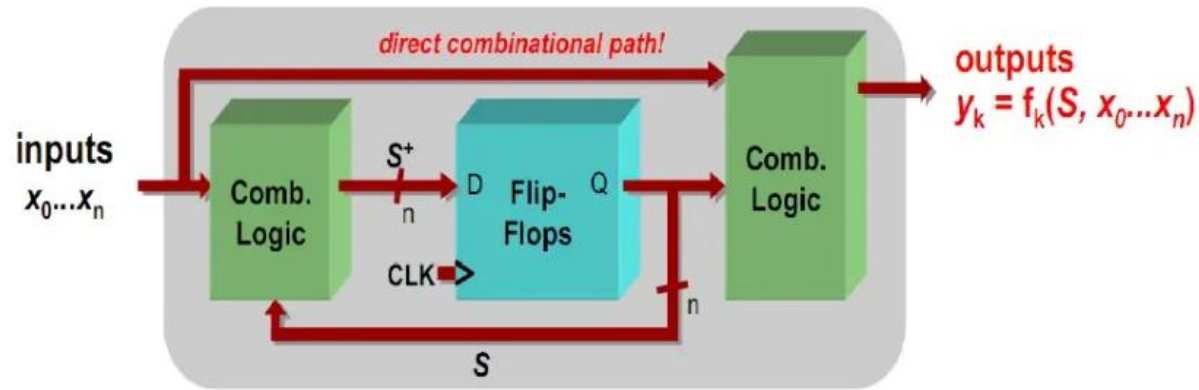
Output is function of present state only



Mealy State Machine:

Output is function of present state and present input

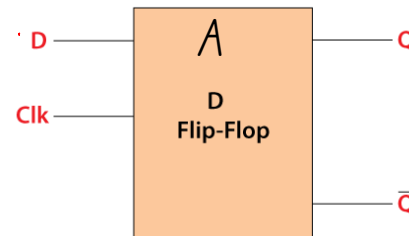
Design example of mealy Machine



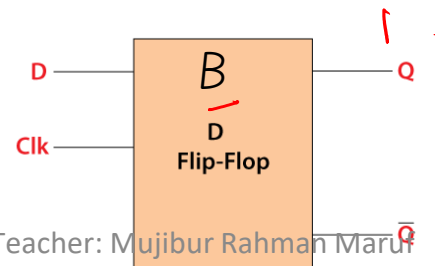
Output is function of present state and present input

D flip flops

X



y

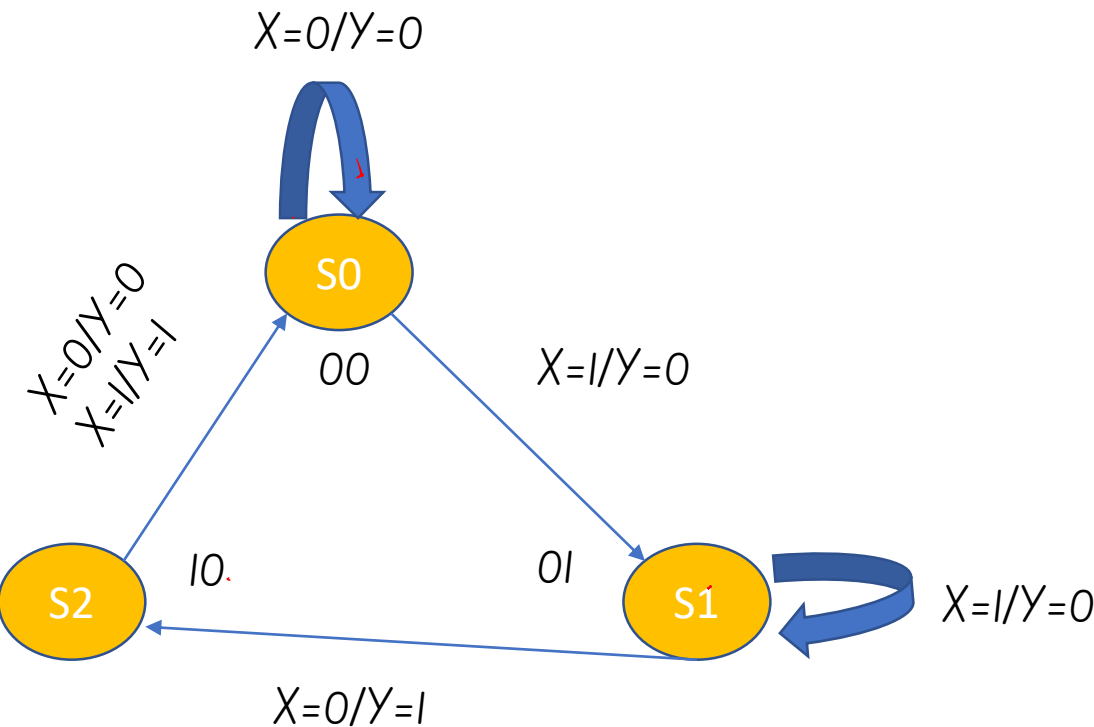


State Diagram

S0=00

S2=01

S3=10



Present State		Input	Next State		Output
A	B	x	A+	B+	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	0	1	0
1	0	0	0	0	0
1	0	1	0	0	1
1	1	0	x	X	x
1	1	1	x	x	x

Design example of mealy Machine

Present State		Input	Next State		Output
A	B	x	A+	B+	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	0	1	0
1	0	0	0	0	0
1	0	1	0	0	1
1	1	0	x	X	x
1	1	1	x	x	x

		BX			
A		00	01	11	10
	0	0	0	0	1
	1	0	0	x	x

$$A+ = BX'$$

		BX			
A		00	01	11	10
	0	0	1	1	0
	1	0	0	x	x

$$B+ = A'X$$

		BX			
A		00	01	11	10
	0	0	0	0	1
	1	0	1	x	x

$$Y = BX' + AX$$

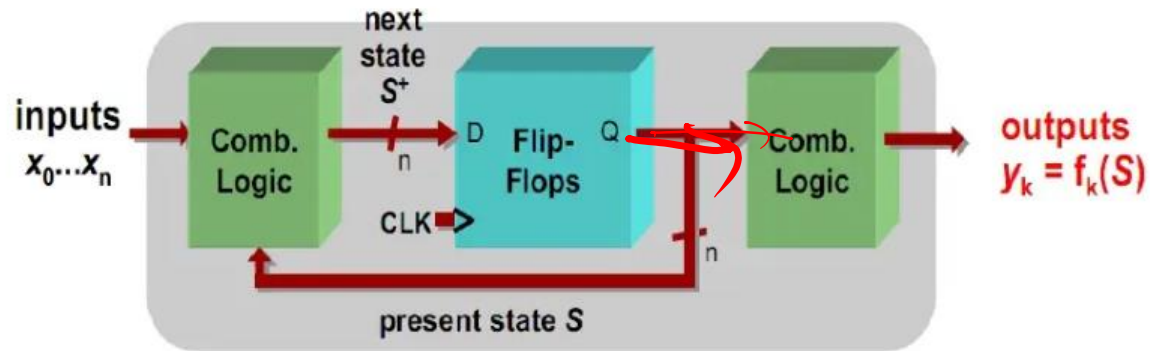
Design example of mealy Machine

Draw the circuit

Class work

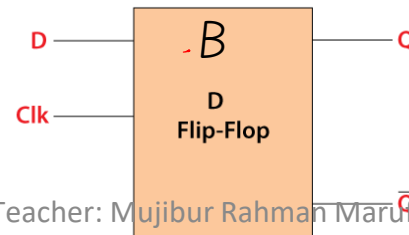
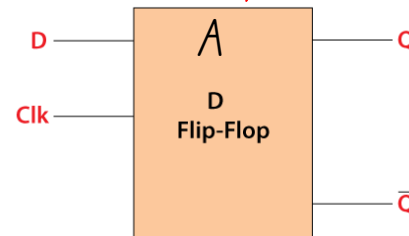
Design example of moore Machine

MOORE STATE MACHINE



Output is function of present state only

X



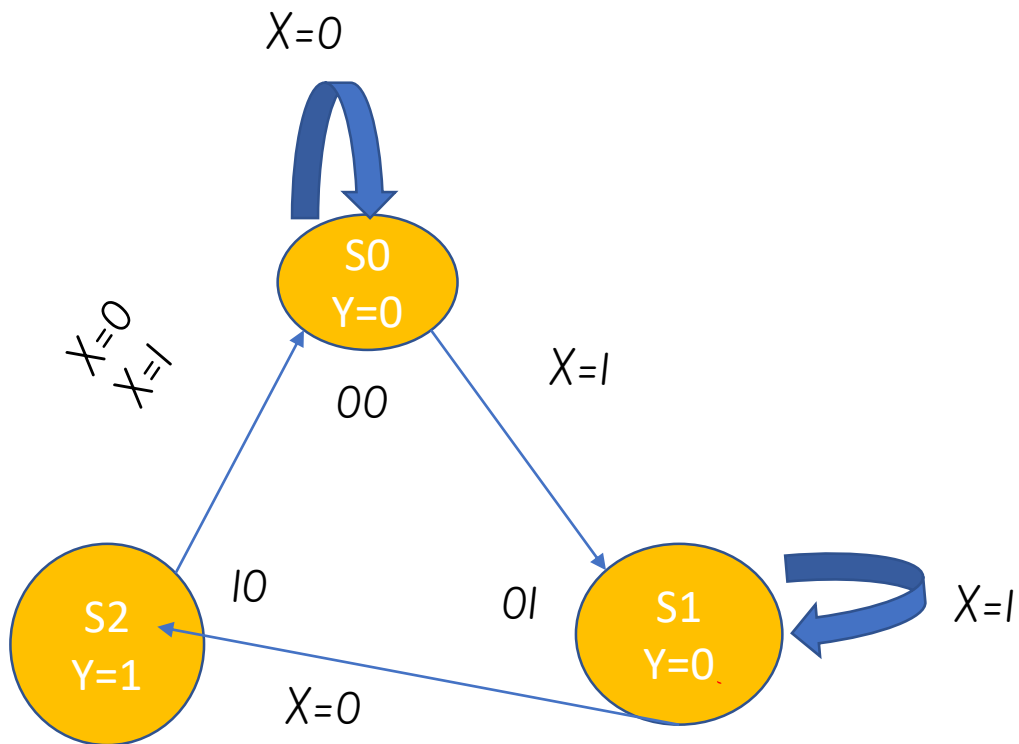
y

State Diagram

$S0=00$

$S1=01$

$S2=10$



Present State		Input	Next State		Output
A	B	x	A+	B+	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	0	0	1
1	0	1	0	0	1
1	1	0	x	x	X
1	1	1	x	x	x

Design example of moore Machine

Present State		Input	Next State		Output
A	B	x	A+	B+	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	0	0	1
1	0	1	0	0	1
1	1	0	x	x	x
1	1	1	x	x	x

		BX			
A		00	01	11	10
	0	0	0	0	1
	1	1	0	x	x

$$A+ = A'BX'$$

		BX			
A		00	01	11	10
	0	0	1	1	0
	1	1	0	x	x

$$B+ = A'X$$

		BX			
A		00	01	11	10
	0	0	0	0	0
	1	1	1	x	x

$$Y = AB'$$

Design example of moore Machine

Draw the circuit

Class work

CSE_2323

Sequence Generator

Sequence Generator

5-bit Sequence Generator: 10011

- The minimum number of flip-flops (n) required to generate a sequence of length N is given by the equation $N < 2^n$.
- Here $N = 5$ and hence, the minimum number of flip-flops required (n) to satisfy the above condition is 3.
- If the given sequence does not lead to five distinct states, then more than three flip-flops may be required.

The given sequence (10011) is listed under Q_2 and the sequence under Q_1 and Q_0 are the same sequence delayed by one and two clock pulses respectively as indicated by arrow marks.

- All 5 states are distinct and hence three flip-flops are sufficient to implement the sequence generator.
- The last column gives the serial input required at the shift register (i.e., D_2 of MSB flip-flop), assuming 3 flip-flops are used and considering the output at Q_2 .
- The K-map for the serial input (Z) is shown in Figure.

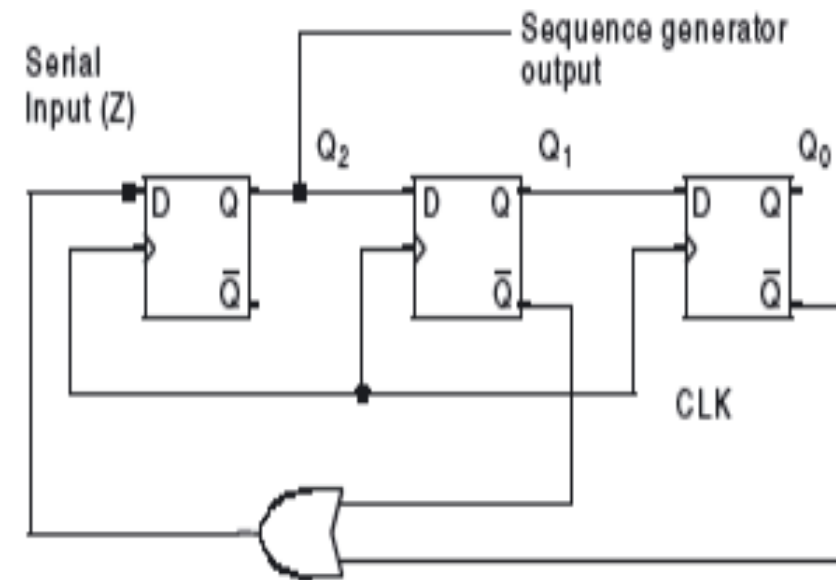
Serial Input	Flip-Flop outputs		
Z	Q_2	Q_1	Q_0
0	1	1	1
0	0	1	1
1	0	0	1
1	1	0	0
1	1	1	0

Sequence Generator

		$Q_1 Q_0$			
		00	01	11	10
Q_2	0	X	1	0	X
	1	1	X	0	1

Q'_1 Q'_0

$$Z = Q'_1 + Q'_0$$



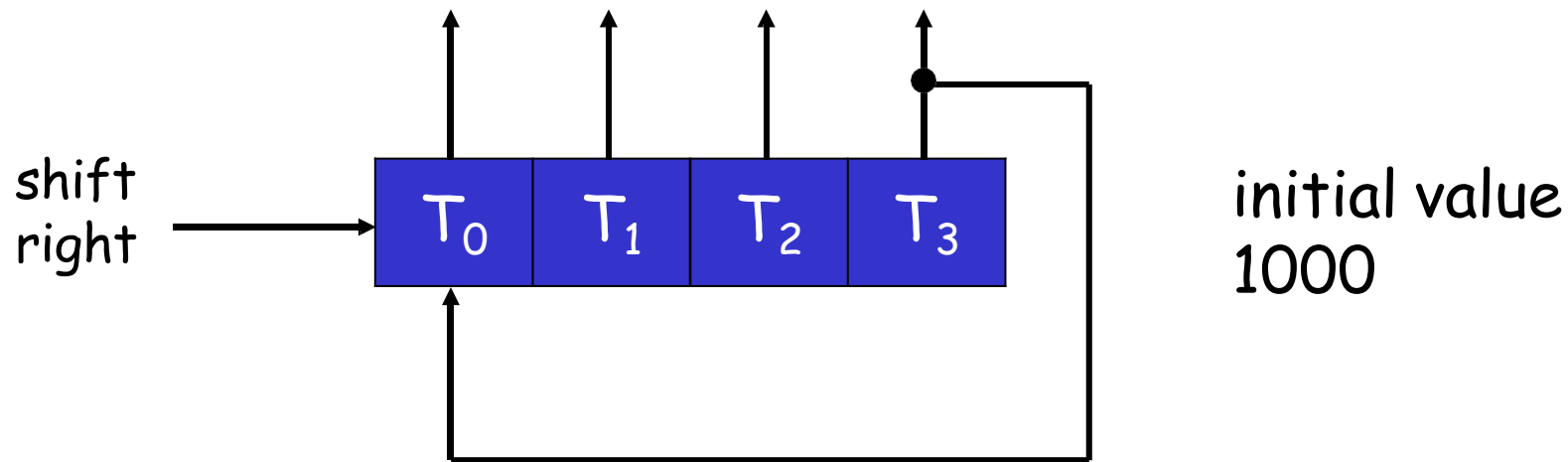
Ring counter

- Ring Counter

- Timing signals control the sequence of operations in a digital system
- A ring counter is a circular shift register with only one flip-flop being set at any particular time, all others are cleared.

almost the same as the shift counter.

The only change is that the output of the last flip-flop is connected to the input of the first flip-flop in the case of the ring counter



4 bit Ring counter

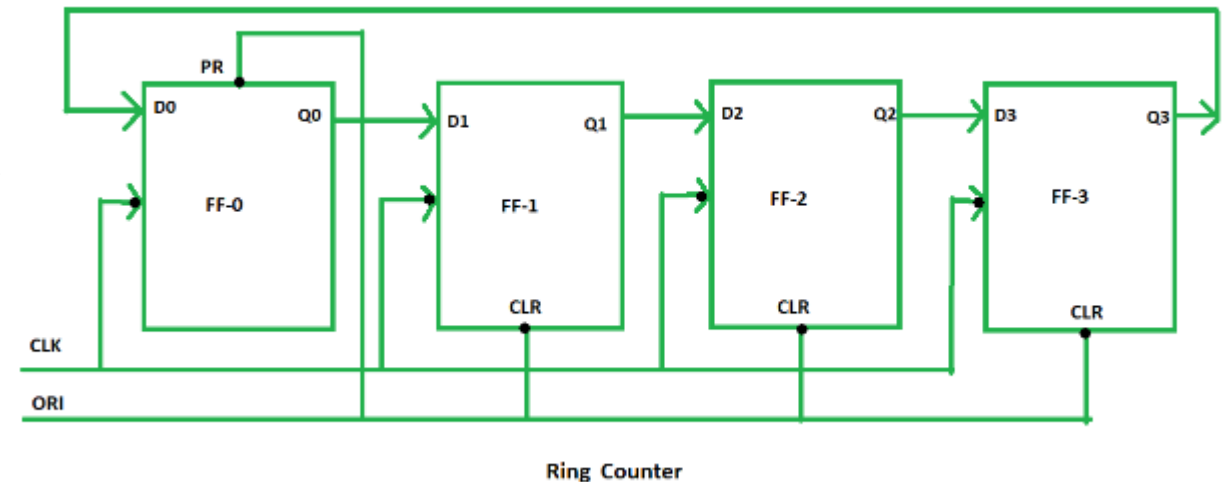
clock pulse (CLK) is applied to all the flip-flops simultaneously.

Therefore, it is a Synchronous Counter.

Overriding input (ORI) for each flip-flop. Preset (PR) and Clear (CLR) are used as ORI.

When PR is 0, then the output is 1. And when CLR is 0, then the output is 0. Both PR and CLR are active low signal that always works in value 0.

ORI is connected to Preset (PR) in FF-0 and it is connected to Clear (CLR) in FF-1, FF-2, and FF-3. Thus, output $Q = 1$ is generated at FF-0, and the rest of the flip-flop generates output $Q = 0$. This output $Q = 1$ at FF-0 is known as Pre-set 1 which is used to form the ring in the Ring Counter.

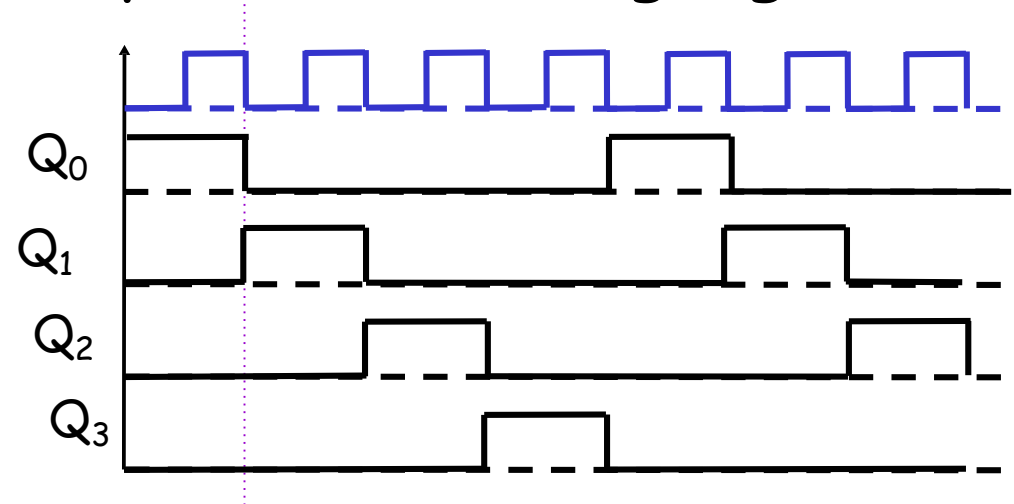


4 bit Ring counter

ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

PRESETED 1

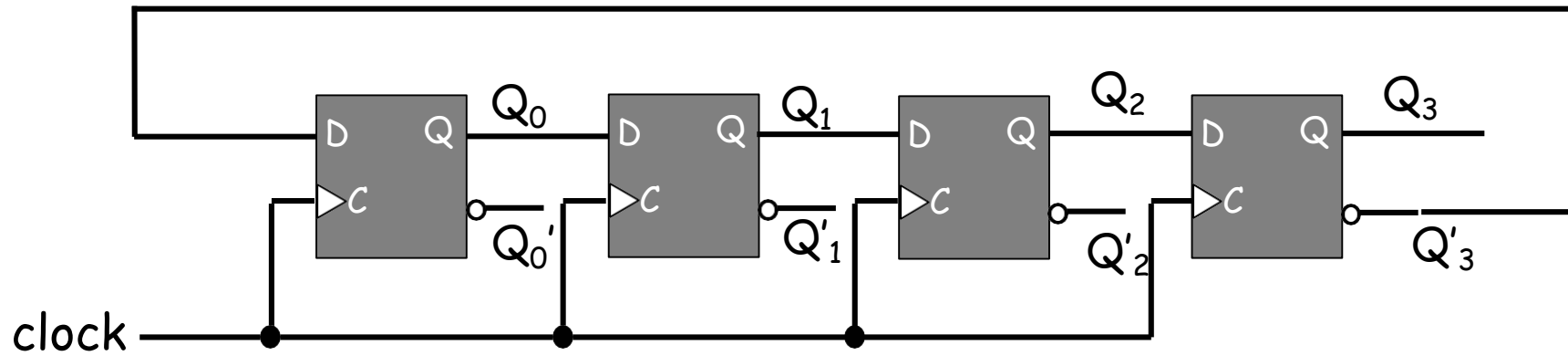
- Sequence of timing signals



at each clock pulse, the preset 1 is shifted to the next flip-flop and
at each clock pulse, the preset 1 is shifted to the next flip-flop and
thus forms a Ring. From the above table, we can say that there are 4
thus forms a Ring. From the above table, we can say that there are 4
states in a 4-bit Ring Counter.
states in a 4-bit Ring Counter.

Johnson counter

- A k -bit ring counter can generate k distinguishable states
- The number of states can be doubled if the shift register is connected as a switch-tail ring counter



Johnson counter

Table 1: Truth table of 4-bit Johnson Counter

State	Q0	Q1	Q2	Q3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0